Introduction to Electronic Design Automation

Homework 2

(Problems 1-5)

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1 Cofactor

1. (a)

The equalities does not hold for u and v is a literal for the same variable. For example, let $f(x) = x$, $u = x$, and $v = \neg x$. Then, $f_u = 1$ and $(f_u)_v = 1$, but $f_v = 0$ and $(f_v)_u = 0$.

1. (b)

Let v be a literal of variable x. If $v = x$, then

$$(\neg f)_v = (\neg f)|_{x=1} = \neg(f|_{x=1}) = \neg(f_v)$$

If $v = \neg x$, then

$$(\neg f)_v = (\neg f)|_{x=0} = \neg(f|_{x=0}) = \neg(f_v)$$

Hence, $(\neg f)_v = \neg(f_v)$ for any literal v.

1. (c)

Let v be a literal of variable x. By Shannon expansion,

$$f = xf_x + \neg xf_{\neg x}$$

and

$$g = xg_x + \neg xg_{\neg x}.$$

We first prove the case $v = x$. Using Shannon expansion on the function $f \oplus g$, we have

$$f \oplus g = x(f \oplus g)_x + \neg x(f \oplus g)_{\neg x}.$$

On the other hand, substituting the Shannon expansions of f and g into $f \oplus g$, we get

$$f \oplus g = (xf_x + \neg xf_{\neg x}) \oplus (xg_x + \neg xg_{\neg x}).$$

When $x = 1$, this becomes

$$(f \oplus g)_v = (f \oplus g)|_{x=1} = f_x \oplus g_x$$

Next, when $v = \neg x$, setting $x = 0$ in the same expansion gives

$$(f \oplus g)_v = (f \oplus g)|_{x=0} = f_{\neg x} \oplus g_{\neg x} = f_v \oplus g_v.$$

Therefore, in both cases,

$$(f \oplus g)_v = f_v \oplus g_v.$$

Hence $(f \oplus g)_v = (f_v) \oplus (g_v)$ is true.

2 Quantified Boolean Formula

3 Boolean Function Representation

4 Application of Quantified Boolean Formula

5 Binary Decision Diagram